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The Editor-in-Chief Journal of King Saud University – Computer and Information Sciences

Dear Editor,

We submit the manuscript **"Duplicate Data Rows Make Prediction Uncertainty Look Better Than It Is: A Simple Check and Fix for New Cases"** for consideration as a regular research article.

**The practical problem.** Teams increasingly attach conformal prediction sets to classifiers and regressors to state how sure a prediction is. They check the promised coverage on a random test split. When tabular data contain repeated rows, as is common in software-defect, network-intrusion, phishing and clinical data, this check is misleading: many test rows have a copy in the training data. On the NASA mc1 defect data, the usual check reports 91.7% coverage for 90% sets, while genuinely new modules are covered only 62.7% of the time and 37% of them receive an empty set.

**What the paper offers.**

- A check that needs only the data. It predicts, before any model is trained, how far coverage of new cases can fall. It includes a new closed-form expression for the share of test rows whose training copies carry the same label.
- A fix: calibrating seen and new cases separately. This is standard Mondrian calibration, applied to the groups that the data imply.
- A reporting rule.
- An open-source tool, *covercheck*, that runs all three steps and writes a one-page report.

**The evidence.**

- **Data and models.** The routine was tested on 12 application data sets, 103 benchmark data sets and controlled experiments with exact and noisy copies. It was also tested on two fresh collections of 20 and 27 data sets, each analysed under a plan committed before its data were downloaded. Learners were random forests, LightGBM and linear models.
- **The check.** It flagged all 13 fresh data sets with a real shortfall (95% CI for sensitivity 75–100%).
- **The fix.** With it, new-case coverage never fell more than 2.5 percentage points below the target.
- **The tool.** It reproduced the research pipeline on 20 data files downloaded afresh.
- **Reproducibility.** Every number in the paper is generated from archived result files.

**Fit with the journal.** The paper addresses a practical, broadly relevant problem in machine-learning evaluation. It is framed around real applications and delivers usable software, which matches the journal's applied computer-science readership.

**Main limitations, stated in the paper.**

- The predicted gap works as a limit and a screen, not as a forecast: on the first fresh collection it did not rank data sets by the size of their shortfall.
- Exact matching misses near-copies; calibration by distance handles them.
- A promise over all new cases does not protect a rare class.
- Several analyses were added after results were seen; they are labelled as such.
- The work was carried out with substantial assistance from an AI system, as declared in the manuscript. The authors take full responsibility for it.

**Statements.** [AUTHOR TO CONFIRM: This manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the submission.] All data sets are public (OpenML and UCI identifiers are listed in the Supplementary Material). The method is available as the

open-source package *covercheck*, and the full pipeline is in a version-controlled repository [AUTHOR TO COMPLETE: DOI or anonymized link].

Sincerely,

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